1) Take two numbers and print the sum of both.

```
class Prog1
{
    public static void main(String[] args)
    {
      int a = 12;
      int b = 34;
      int c = a + b;
      System.out.println("SUM OF TWO NUMBER : "+c);
    }
}
```

2) Take a number as input and print the multiplication table for it.

```
import java.util.Scanner;
class Prog2
{
    public static void main (String[] args)
    {
    Scanner scn=new Scanner(System.in);
    System.out.println("ENTER WHICH TABLE YOU WANT : ");
    int n=scn.nextInt();
    System.out.println("ENTER END");
    int end=scn.nextInt();
    for(int i=1;i<=end;i++)
    {
      System.out.println(i+"*"+n+" = "+i*n);
    }
      scn.close();
    }
}
```

OUTPUT

```
ENTER WHICH TABLE YOU WANT :
6
ENTER END
5
1*6 = 6
2*6 = 12
3*6 = 18
4*6 = 24
5*6 = 30
```

3) Input a year and find whether it is a leap year or not.

```java
import java.util.Scanner;
class Prog3
{
  public static void main(String[] args)
  {
    Scanner scn=new Scanner(System.in);
    System.out.println("ENTER YEAR : ");
    int year=scn.nextInt();
   if(year%4==0 && year%100!=0 || year%400==0)
     {
       System.out.println("THIS IS A LEAP YEAR");
 }
  else
    {
      System.out.println("THIS IS NOT A LEAP YEAR");
    }
   scn.close();
   }
 }
```

OUTPUT

```
ENTER YEAR :
2014
THIS IS NOT A LEAP YEAR
PS C:\Users\ArunachalamP
ENTER YEAR :
2000
THIS IS A LEAP YEAR
```

CONCEPT

1. What is a Leap Year?

- A leap year is a year that has **366 days** instead of 365 days.
- An extra day is added to **February**.
- February has **29 days** in a leap year.

2. Why Do We Need a Leap Year?

- The Earth takes approximately **365.2422 days** to complete one revolution around the Sun.
- A normal calendar year has only **365 days**.
- The extra **0.2422 day** accumulates every year.
- To adjust this difference, one extra day is added every four years.

3. Basic Rule

- If a year is divisible by **4**, it is generally a leap year.

Examples:

- 2024 ÷ 4 = 506 → Leap Year
- 2028 ÷ 4 = 507 → Leap Year

4. Century Year

- A **century year** is a year ending with **00**.
- Examples: **1700, 1800, 1900, 2000, 2100**

5. Rule for Century Years

- Not all century years are leap years.
- A century year must be divisible by **400** to be a leap year.

Examples:

- 1900 ÷ 400 = 4.75 → Not a Leap Year
- 2000 ÷ 400 = 5 → Leap Year
- 2100 ÷ 400 = 5.25 → Not a Leap Year
- 2400 ÷ 400 = 6 → Leap Year

6. Final Rules for Leap Year

A year is a leap year if:

☑ It is divisible by **4** and not divisible by **100**

OR

☑ It is divisible by **400**

4. Take 2 numbers as inputs and find their HCF and LCM.

```
import java.util.Scanner;

class Prog4

{

  public static void main(String[] args)

   {

    Scanner scn=new Scanner(System.in);

    System.out.println("ENTER NUM!:");

    int num1=scn.nextInt();

    System.out.println("ENTER NUM2:");

    int num2=scn.nextInt();

    int hfc= 0;

   for(int i=1;i<=num1&&i<=num2;i++)

  {

     if(num1%i==0 && num2%i==0)
```

```
        {
              hfc=i;
           }
          }
        System.out.println("HCF: " + hfc);
        int lcm=num1*num2/hfc;
        System.out.println("LCM: " + lcm);
        scn.close();
        }
    }
```

OUTPUT

```
ENTER NUM!:
16
ENTER NUM2:
24
HCF: 8
LCM: 48
```

CONCEPT

1) HCF (Highest Common Factor)

HCF means **the biggest number that divides both numbers exactly**.

Example:
For **12 and 18**

Factors of 12 → 1, 2, 3, 4, 6, 12
Factors of 18 → 1, 2, 3, 6, 9, 18

Common factors → 1, 2, 3, 6

Highest common factor = **6**

LCM Formula

LCM=(num1×num2)/HCF

Example:

For 12 and 18:

$$
\begin{aligned}
LCM &= (12 \times 18)/6 \\
&= 216/6 \\
&= 36
\end{aligned}
$$